Foundations of Computing

Demo slides for the Defter reader

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Agenda

Part 1
Logic and Propositions

Part 2
Sets and Functions

Part 3
Algorithms and Complexity

Part 4
Data Structures

Part 5
Graphs

Part 6
Finite Automata

Part 1

Logic and Propositions

Propositions - Connectives - Truth tables - Implication

Logic and Propositions

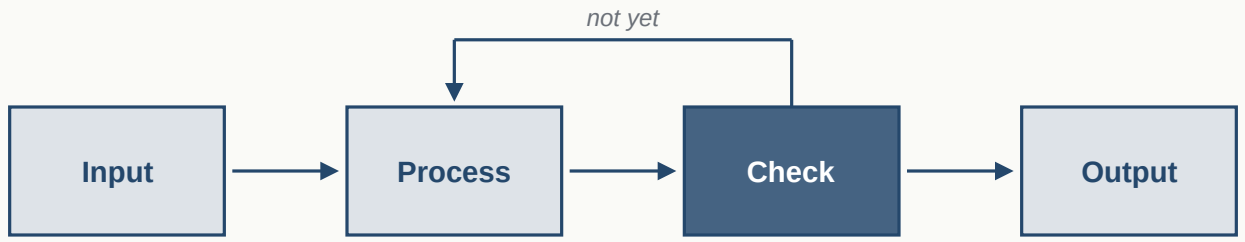
- The basic connectives are negation (not), conjunction (and), disjunction (or) and implication (if ... then).
- Implication is the connective that surprises most beginners. 'If p then q' is false in exactly one case: p is true and q is false.
- Laws such as De Morgan's rules let us rewrite formulas: not (p and q) is the same as (not p) or (not q).

p	q	p and q	p or q	if p then q
T	T	T	T	T
T	F	F	T	F
F	T	F	T	T
F	F	F	F	T

Truth table of the basic connectives

Logic and Propositions: key idea

A truth table is a complete definition: nothing about a connective is left open.



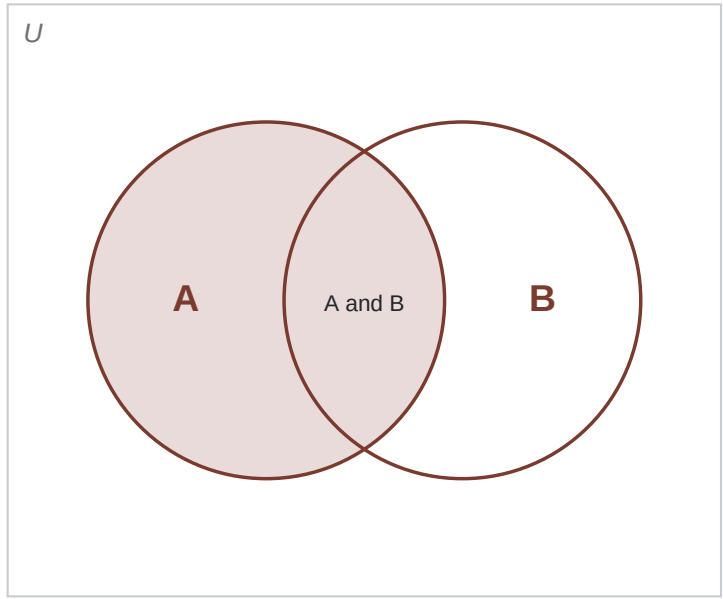
Part 2

Sets and Functions

Sets and elements - Describing sets - Set operations - Venn diagrams

Sets and Functions

- Union, intersection and difference build new sets from old ones.
- A function f from A to B assigns exactly one element of B to every element of A. A is the domain and B is the codomain.
- A function that is both injective and surjective is a bijection.



Two sets inside a universe U

Sets and Functions: key idea

A function gives every input exactly one output; a bijection can also be reversed.

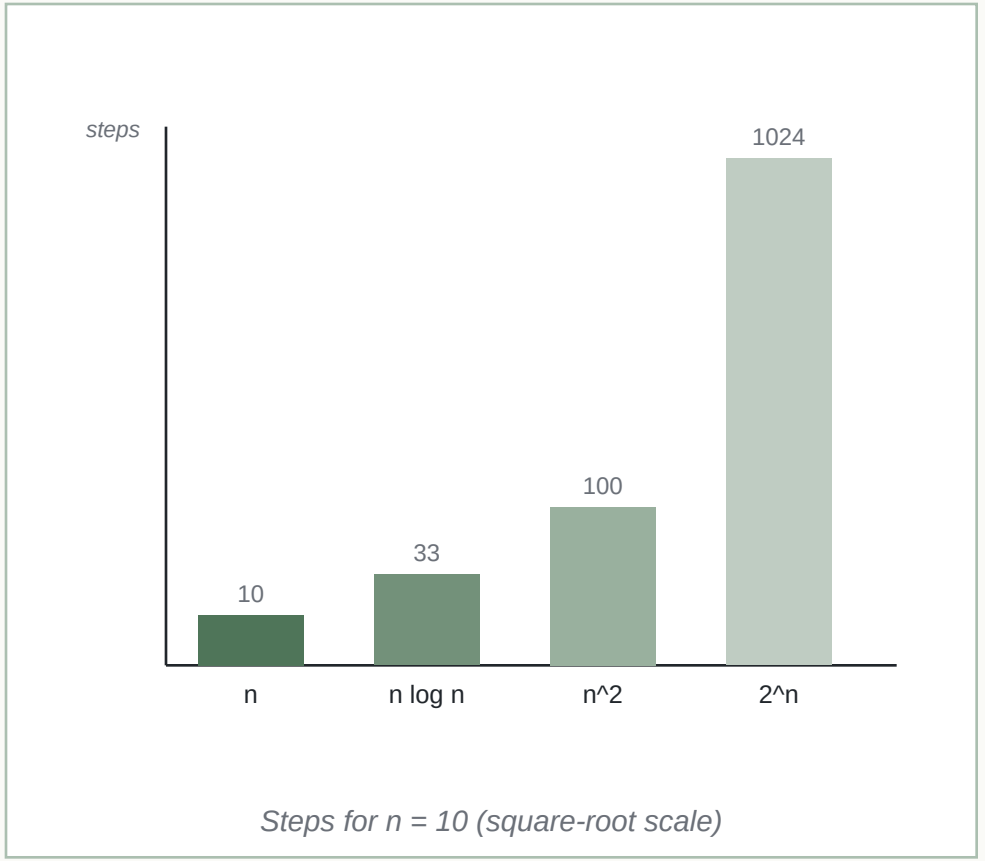
Part 3

Algorithms and Complexity

What is an algorithm - Counting steps - Big-O notation - Linear search

Algorithms and Complexity

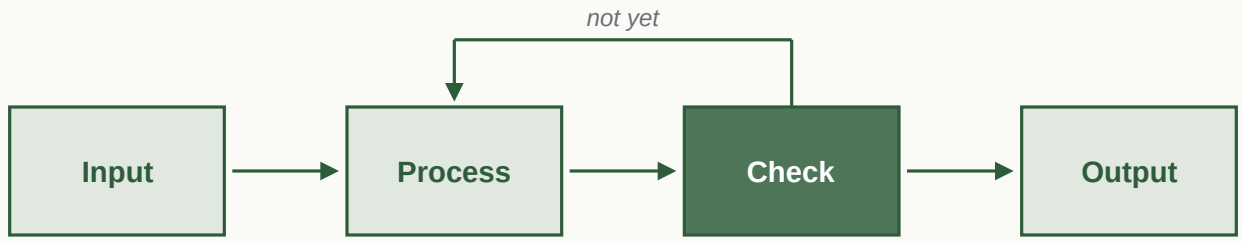
- To compare algorithms we count basic steps as a function of the input size n.
- Linear search looks at the elements one by one and needs up to n comparisons.
- Sorting is one of the most studied problems.



Steps for n = 10 (square-root scale)

Algorithms and Complexity: key idea

For large inputs, the growth rate of an algorithm matters more than the speed of the machine.



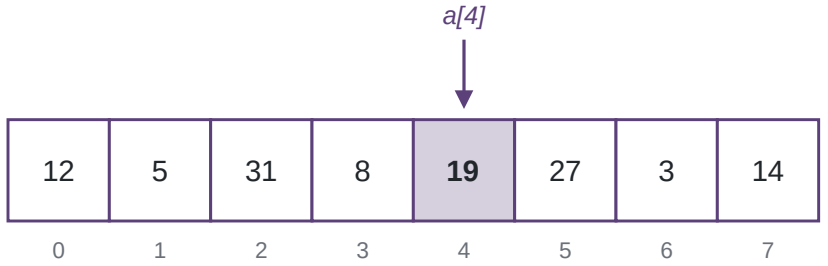
Part 4

Data Structures

Choosing a structure - Arrays - Linked lists - Stacks

Data Structures

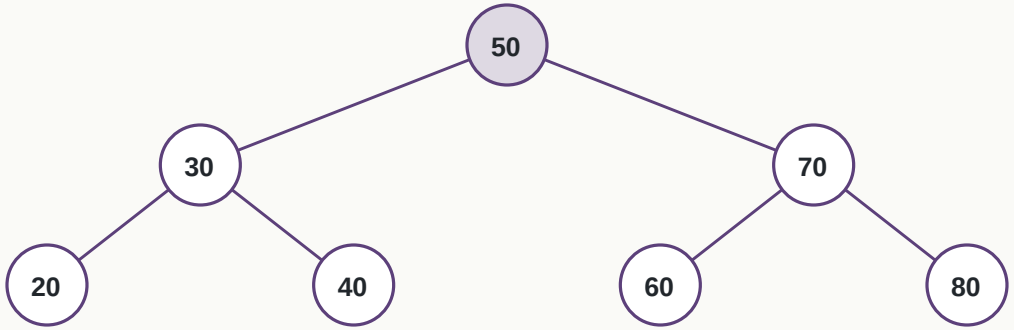
- An array stores elements next to each other in memory.
- Stacks and queues restrict how we access the data.
- A hash table maps keys to values.



An array with its indices

Data Structures: key idea

Pick the structure from the questions you ask most often, not from habit.



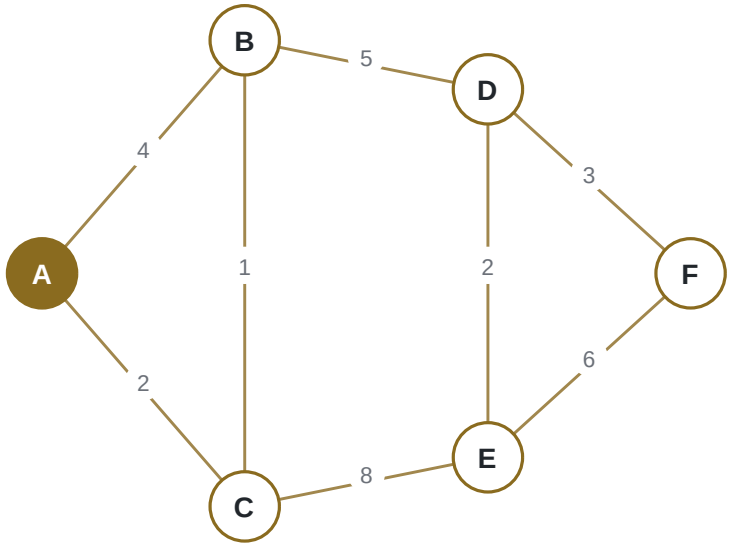
Part 5

Graphs

Vertices and edges - Directed and weighted graphs - Paths and cycles - Breadth-first search

Graphs

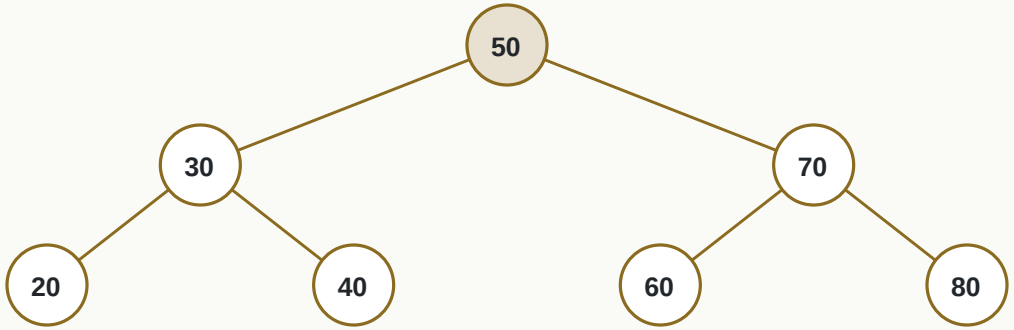
- Edges can be directed, like one-way streets, or undirected.
- Breadth-first search explores a graph layer by layer: first all neighbours of the start vertex, then their neighbours, and so on.
- Dijkstra's algorithm finds shortest paths when edges have non-negative weights.



A weighted graph with six vertices

Graphs: key idea

Many different problems become the same problem once you see the graph behind them.



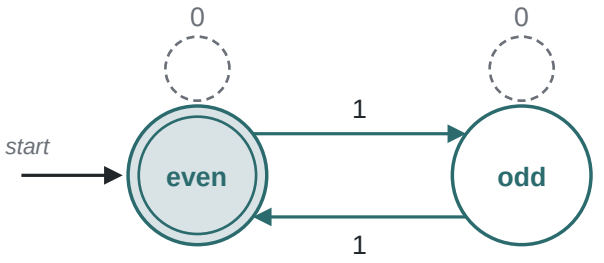
Part 6

Finite Automata

States and transitions - Accepting words - Languages - An example automaton

Finite Automata

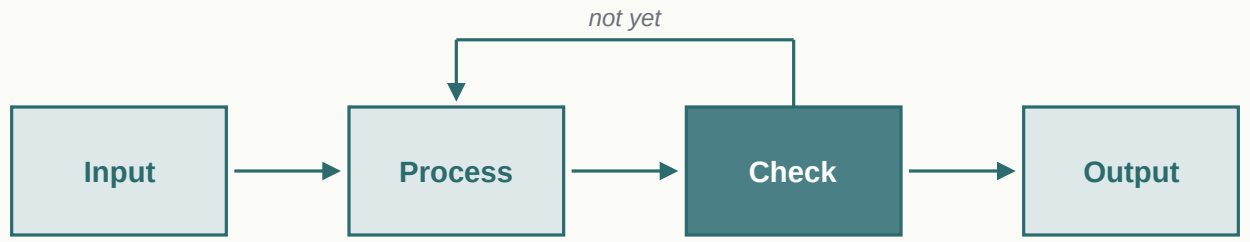
- The set of all words an automaton accepts is its language.
- Regular expressions describe the same languages as finite automata.
- Not every language is regular.



Automaton for an even number of ones

Finite Automata: key idea

A finite automaton remembers only its current state, so it cannot count without limit.



Exercises

- Write the truth table of (p and not q) or (not p and q). Which connective from the chapter does it describe?
- Give a function from {1, 2, 3} to {a, b} that is surjective. Can such a function be injective? Explain.
- Which data structure would you use for the undo history of a text editor? Give a reason.
- Run breadth-first search on your graph from exercise 9 and list the vertices in the order they are visited.

Summary

Logic and Propositions

A truth table is a complete definition: nothing about a connective is left open.

Sets and Functions

A function gives every input exactly one output; a bijection can also be reversed.

Algorithms and Complexity

For large inputs, the growth rate of an algorithm matters more than the speed of the machine.

Data Structures

Pick the structure from the questions you ask most often, not from habit.

Graphs

Many different problems become the same problem once you see the graph behind them.

Finite Automata

A finite automaton remembers only its current state, so it cannot count without limit.

About these slides

- Generated by scripts/demo-pdf-uret.mjs: the script writes the PDF directly, with no extra packages.
- Landscape 16:9 pages test how the reader fits wide pages into book mode and PDF mode.
- All text and figures were written for this demo. Nothing is copied from real lecture material.

Thank you

End of the demo slides - 24 / 24